

## CRANE LAKE SCOTTS SPB, MN, USA

WMO: 727473

Lat: 48.267N

Lon: 92.487W

Elev: 1119

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94977

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.2      | -21.8 | -35.0                         | 0.8 | -27.4 | -27.2 | 1.3 | -21.3 | 14.7                  | 14.5 | 12.7 | 14.0 | 0.1                   | 160  | 0.548 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.5                   | 85.7            | 69.3 | 82.1 | 67.4 | 80.5 | 66.4 | 73.5                | 81.5 | 71.2 | 79.1 | 68.9 | 76.0 | 5.8                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.7                            | 122.1 | 78.2 | 68.4 | 108.7 | 75.4 | 66.0 | 100.0 | 73.3 | 37.8          | 81.8 | 35.6 | 79.0 | 33.7 | 76.5 | 84.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|--|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |  |
| 13.3              | 11.7 | 10.3 | DB                         | -33.1 | 90.8               | 6.7 | 3.8                                                | -38.0 | 93.5       | -41.9 | 95.8       | -45.6 | 97.9       | -50.5 | 100.7 |  |
|                   |      |      | WB                         | -33.3 | 77.4               | 6.5 | 2.9                                                | -38.0 | 79.4       | -41.8 | 81.1       | -45.4 | 82.7       | -50.1 | 84.8  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 38.1   | 6.7   | 9.7   | 23.6  | 37.5  | 50.5  | 60.8  | 65.3  | 63.1  | 55.5  | 42.3  | 27.7  | 13.0  |
|                                                                               | DBStd       | 23.13  | 14.79 | 14.10 | 13.49 | 9.68  | 8.49  | 7.15  | 5.89  | 6.42  | 8.33  | 8.71  | 10.92 | 13.63 |
|                                                                               | HDD50       | 5908   | 1343  | 1130  | 819   | 387   | 98    | 5     | 0     | 0     | 37    | 272   | 669   | 1148  |
|                                                                               | HDD65       | 10005  | 1808  | 1550  | 1282  | 824   | 456   | 164   | 70    | 114   | 303   | 703   | 1118  | 1613  |
|                                                                               | CDD50       | 1573   | 0     | 0     | 2     | 13    | 113   | 329   | 473   | 406   | 201   | 35    | 1     | 0     |
|                                                                               | CDD65       | 193    | 0     | 0     | 0     | 0     | 6     | 37    | 78    | 54    | 17    | 1     | 0     | 0     |
|                                                                               | CDH74       | 2391   | 0     | 0     | 0     | 8     | 151   | 484   | 915   | 626   | 194   | 13    | 0     | 0     |
|                                                                               | CDH80       | 587    | 0     | 0     | 0     | 1     | 36    | 124   | 232   | 154   | 38    | 2     | 0     | 0     |
| Wind                                                                          | WSAvg       | 3.3    | 2.9   | 3.1   | 3.5   | 3.8   | 3.8   | 3.0   | 2.8   | 2.7   | 3.3   | 3.6   | 3.6   | 2.9   |
| Precipitation                                                                 | PrecAvg     | 27.8   | 0.9   | 0.7   | 1.1   | 2.0   | 3.0   | 4.0   | 4.0   | 3.2   | 3.7   | 2.6   | 1.5   | 1.1   |
|                                                                               | PrecMax     | 33.6   | 2.0   | 1.3   | 3.3   | 4.4   | 5.1   | 11.1  | 8.0   | 6.2   | 7.4   | 4.3   | 3.4   | 2.3   |
|                                                                               | PrecMin     | 23.1   | 0.4   | 0.1   | 0.4   | 0.6   | 1.1   | 1.7   | 1.7   | 0.7   | 1.2   | 0.6   | 0.1   | 0.6   |
|                                                                               | PrecStd     | 3.1    | 0.4   | 0.4   | 0.7   | 0.9   | 1.1   | 2.0   | 1.9   | 1.5   | 1.8   | 1.1   | 0.9   | 0.5   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.3  | 45.4  | 62.9  | 72.6  | 84.2  | 90.0  | 90.4  | 89.7  | 84.2  | 74.9  | 58.7  | 44.8  |
|                                                                               |             | MCWB   | 35.2  | 37.9  | 50.4  | 54.3  | 62.4  | 72.8  | 73.0  | 73.4  | 67.9  | 59.1  | 50.5  | 37.3  |
|                                                                               | 2%          | DB     | 35.8  | 39.2  | 53.9  | 66.0  | 78.8  | 82.5  | 84.4  | 83.9  | 79.5  | 68.0  | 52.4  | 37.2  |
|                                                                               |             | MCWB   | 33.4  | 34.9  | 44.9  | 49.7  | 59.1  | 66.7  | 69.7  | 69.3  | 66.0  | 55.8  | 46.5  | 34.3  |
|                                                                               | 5%          | DB     | 31.7  | 35.9  | 47.3  | 61.2  | 72.7  | 79.4  | 81.9  | 81.0  | 74.9  | 62.9  | 46.4  | 34.6  |
|                                                                               |             | MCWB   | 30.4  | 33.2  | 40.2  | 47.2  | 56.7  | 64.4  | 68.2  | 67.8  | 63.8  | 54.2  | 41.8  | 32.9  |
|                                                                               | 10%         | DB     | 27.9  | 31.7  | 42.7  | 55.1  | 68.3  | 75.3  | 79.4  | 77.2  | 71.8  | 56.7  | 43.2  | 31.7  |
|                                                                               |             | MCWB   | 26.4  | 30.2  | 36.6  | 43.9  | 55.3  | 62.7  | 66.9  | 65.2  | 61.9  | 50.2  | 39.2  | 30.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 36.5  | 39.7  | 54.0  | 57.7  | 68.1  | 74.3  | 76.3  | 76.5  | 71.6  | 63.7  | 53.8  | 38.2  |
|                                                                               |             | MCDB   | 39.4  | 43.5  | 57.7  | 68.9  | 74.8  | 83.9  | 84.1  | 84.3  | 80.4  | 68.6  | 56.3  | 42.5  |
|                                                                               | 2%          | WB     | 33.3  | 35.8  | 45.8  | 52.3  | 63.4  | 71.1  | 73.7  | 73.0  | 68.7  | 58.9  | 46.3  | 35.2  |
|                                                                               |             | MCDB   | 35.0  | 38.9  | 52.5  | 62.4  | 71.5  | 80.0  | 81.2  | 80.8  | 76.3  | 66.0  | 50.2  | 37.3  |
|                                                                               | 5%          | WB     | 30.1  | 33.0  | 40.8  | 48.6  | 60.1  | 68.4  | 71.4  | 70.0  | 66.2  | 54.6  | 43.0  | 32.9  |
|                                                                               |             | MCDB   | 31.4  | 35.6  | 47.0  | 59.4  | 69.4  | 75.3  | 78.9  | 77.4  | 71.8  | 61.2  | 46.6  | 34.7  |
|                                                                               | 10%         | WB     | 26.7  | 29.5  | 36.9  | 45.2  | 56.8  | 65.9  | 69.0  | 67.5  | 63.5  | 51.4  | 39.0  | 30.2  |
|                                                                               |             | MCDB   | 27.9  | 31.5  | 41.1  | 54.0  | 65.0  | 72.3  | 76.0  | 74.4  | 68.7  | 55.9  | 42.8  | 31.5  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 21.0  | 24.5  | 23.8  | 23.5  | 24.5  | 23.3  | 23.5  | 23.4  | 22.3  | 17.7  | 14.6  | 16.9  |
|                                                                               |             | MCDBR  | 19.0  | 23.5  | 27.1  | 33.6  | 33.5  | 28.1  | 26.9  | 26.9  | 28.2  | 27.5  | 21.1  | 18.0  |
|                                                                               |             | MCWBR  | 16.5  | 18.7  | 17.8  | 19.4  | 17.4  | 14.7  | 14.5  | 14.5  | 16.1  | 16.5  | 15.0  | 14.8  |
|                                                                               | 5% WB       | MCDBR  | 17.3  | 20.5  | 25.0  | 29.8  | 27.4  | 22.8  | 24.0  | 23.7  | 22.8  | 23.7  | 19.7  | 17.0  |
| MCWBR                                                                         |             | 15.4   | 16.7  | 17.3  | 18.6  | 16.3  | 14.0  | 15.0  | 14.4  | 14.7  | 15.6  | 15.0  | 14.3  |       |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.257  | 0.270 | 0.321 | 0.337 | 0.358 | 0.375 | 0.389 | 0.372 | 0.365 | 0.308 | 0.287 | 0.262 |       |
|                                                                               | taud        | 2.347  | 2.316 | 2.325 | 2.419 | 2.412 | 2.387 | 2.371 | 2.448 | 2.437 | 2.584 | 2.508 | 2.390 |       |
|                                                                               | Ebn at Noon | 262    | 284   | 284   | 290   | 287   | 281   | 275   | 276   | 266   | 270   | 251   | 246   |       |
|                                                                               | Edh at Noon | 22     | 28    | 33    | 34    | 36    | 36    | 37    | 33    | 30    | 22    | 19    | 18    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 421    | 797   | 1184  | 1504  | 1667  | 1768  | 1877  | 1590  | 1174  | 685   | 382   | 298   |       |
|                                                                               | RadStd      | 54     | 81    | 110   | 140   | 113   | 139   | 101   | 112   | 96    | 94    | 49    | 30    |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page